Early Churn Detection and Segmentation of Telecom Customers Using Machine Learning

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***Abstract*—Customer churn is a serious issue that telecom companies face because by losing subscribers, particularly during their early days of service, telecom companies may lose a lot of revenue and their customer lifetime value. The proposed study addresses the issue of early churn prediction, which is the loss of customers in the first three months of service, and implements a hybrid of supervised machine learning and customer segmentation. The data set will consist of 7043 customers, demographic, service subscription, and billing data. Extensive preprocessing methods were used such as missing values, categorical variables, numeric features scaling, and class imbalance (SMOTE). Several predictive models, such as Logistic Regression, Random Forest, XGBoost, and LightGBM, were tested on the basis of precision, recall, F1-score, and ROC-AUC. LightGBM performed better (F1-score: 0.97, ROC-AUC: 0.995), and threshold optimization enhanced the impact of predicted retention and minimized possible business expenses. The SHAP values of feature importances gave information on the factors that have the most significant effect on early churn: tenure, monthly charges, and contract type. The KMeans clustering was also used to segment early churners, which identified different groups of behavioral and revenue-based segments, which could be targeted with specific retention strategies. It is a comprehensive approach, a combination of predictive modelling and client segmentation to provide actionable information to help telecom corporations to rationalize retention efforts, minimize loss of revenue and develop information based customer engagement strategies.**

Keywords—Early Churn Prediction, Telecom Churn, Supervised Learning, LightGBM, SMOTE, Customer, ROC-AUC

I. Introduction

Customer churn has emerged as a significant issue to companies that are involved in very competitive business environments especially in the telecommunications industry. The presence of various service providers in the market that sell similar products at similar prices makes it sometimes expensive to attract new customers than to retain the ones that are already there. Customer churn is the situation when subscribers cease their services, which results in the immediate loss of revenue and lower customer lifetime value. Among the various types of churn, early churn, in which customers drop off the first few months of subscription, is particularly of concern because it signifies a lack of satisfaction in the onboarding stage and leads to a low payback on the customer acquisition efforts.

Traditional methods to churn analysis tend to use descriptive statistics or rule systems that might not be able to identify complex trends in customer behavior. As access to customer usage, billing, and demographic data has increased, machine learning (ML) algorithms have become an effective predictor of churn. Supervised learning models have the capability to learn patterns based on historical data and identify the high-churning customers, which allows the organization to implement proactive retention strategies.

This paper is dedicated to the prediction of early customer churn in the field of telecoms with the help of supervised machine learning. This analysis is performed using a publicly available telecom dataset that contains 7043 records of customers that have demographic, service subscription, and billing attributes. The Early Churn is a new target variable that is used to determine a customer who discontinues services within the first three months. Preprocessing stages involve missing value processing, categorical variable encoding, data scaling, and the class imbalance problem, which is solved with the help of the Synthetic Minority Over-sampling Technique (SMOTE).

Several machine learning models are trained and tested with the standard performance measures, which are precision, recall, F1-score, and area under the ROC curve (ROC-AUC). Besides predictive modeling, customer segmentation is also conducted through K-Means clustering to understand the behavior pattern of early churners. The feature importance analysis is also performed to increase the interpretability of the models.

The three top contributions of this paper are as follows: (1) the creation of a correct early churn prediction model with the help of ensemble learning methods, (2) the maximization of decision thresholds in order to minimize possible business expenses, and (3) the division of early churners to give viable insights on specific retention strategies. The suggested strategy illustrates how machine learning could be useful in promoting data-driven decision-making and retention of customers in the telecommunication sector.

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II. LITERATURE REVIEW

The churn prediction of customers is an important research concern in the telecommunications industry, given that it directly affects the business revenue and customer retention policies. Churn, the process of a customer terminating the utilization of a product or service, may lead to serious financial losses in the service provider, as it is usually much costlier to acquire new clients than to keep the current ones [1].Initial research focused mainly on conventional statistical methods, including logistic regression and discriminant analysis. These techniques were not highly effective in non-linear relationships of large customer data that necessitated prediction, although they had a certain predictive power [2], [3].

The introduction of machine learning (ML) practices became a major breakthrough in the research of churn prediction. One of the first types of models to represent complex trends in customer behavior was neural network-based models, in particular, Multi-Layer Perceptron (MLP) networks [2], [3]. These models showed better predictive capability than the traditional statistical models, but had weaknesses of being difficult to interpret, and therefore difficult to use in practice. Also, neural networks were very resource-intensive in terms of computation and hyperparameter optimization, and this limitation also limited their practicality in real-time.

To address these constraints, scholars considered more interpretable and efficient ML models. Algorithms based on decision trees, such as Random Forest (RF), became a popular choice because they can produce rules that are easy to understand and yet have a high level of predictive accuracy [4]. Ensemble-based learning algorithms like RF and Stochastic Gradient Boosting (SGB) have also been shown to be superior over individual classification models by integrating several weak learners to create a strong predictive model [5], [6]. Such approaches are applicable to model intricate customer-related interactions and can address unbalanced data, a usual issue in churn prediction, as non-churners exceed churners [5], [10].

It has been emphasized that feature selection and preprocessing play vital roles in the development of efficient churn prediction models. Other studies have pointed out that proper preprocessing procedures such as dealing with missing data, encoding nominal variables, scaling numerical variables and dealing with an imbalanced number of classes are very important to improve the performance of the models. Chi-square feature selection and Principal Component Analysis (PCA) techniques help to eliminate irrelevant or redundant features, improve computational efficiency, and improve the applicability of models to unseen data, in general [8], [9]. The influence of feature engineering is especially significant in ensemble ML models where well-chosen features have a direct effect on the increased predictive accuracy.

Some studies were also dedicated to the comparison and evaluation of the performance of various algorithms of ML in terms of churn prediction. Random Forest and SGB regularly proved to be the most successful, whereas K-Nearest Neighbors (KNN) and Logistic Regression demonstrated the average accuracy [1], [5], [9]. It is continually showed in literature that ensemble techniques are more effective than traditional classifiers because they can minimize overfitting and effectively address high-dimensional feature spaces [5], [6]. Also, the hybrid models which integrate clustering and classification or multiple classifiers are suggested to further improve predictive performance at a lower cost of computation [6], [7].

Although these have been made, there are a number of challenges that exist in research. The use of accuracy as the main measure of evaluation is one of the weaknesses. Accuracy is a useful measure that represents the overall correctness of predictions, but it cannot entirely represent business impact, including customer lifetime value, retention cost, or profit obtained through proactive retention strategies [10], [11]. The other problem is that the models are not generalizable to other datasets and sectors. It is common that many studies can train models at high performance on particular datasets, and they fail to generalize when these models are applied to new or heterogeneous datasets [9]. This highlights the need to have strong model validation and constant retraining in practical contexts.

The need to have interpretability is another research challenge. Although ensemble models are very accurate in predicting customers, it is important to understand why a customer is likely to churn to develop special retention strategies. Decision trees provide a certain degree of interpretability, yet more sophisticated ensemble models may be a black box and it may be challenging to execute actionable strategies by decision-makers [5], [10]. Moreover, the time-varying behavior of customers, which is shaped by the competition in the market, promotions, or changing preferences of the services, also causes concept drift, the predictive patterns taught by the model can become obsolete with time [1], [10].

To overcome the limitations mentioned above, this paper will only consider classical machine learning techniques that are computationally efficient, interpretable, and applicable to telecom data. As opposed to the methods of deep learning, which demand substantial data sets and intricate computing facilities [12], the specified ML models can be practically implemented in the real world and deliver practical information on customer retention. This study will establish models that are predictive but at the same time interpretable and easy to apply by systematically comparing various ML algorithms using common telecom data.

The gap in the research that was determined through literature may be summarised as follows: (1) numerous studies do not include business-relevant measures other than accuracy, including profit or customer lifetime value. (2) The generalizability of the model across datasets is usually overlooked and restricts real-world use. (3) Ensemble models are often black-box models that are accurate but not interpretable to produce actionable business strategies. (4) The concept drift and dynamic customer behavior are rarely discussed, and it lowers the predictive model reliability in the long-term perspective [1], [5], [9], [10]. Through these gaps, this research will be useful in the development of effective, strong, and interpretable machine learning models to predict telecom churn.

III. METHODOLOGY

A. Data Collection

The data employed in the present research is the publicly available WA_Fn-UseC_-Telco-Customer-Churn.csv, comprising of 7043 customer records and 21 attributes of demographic data, account-related data, and service usage patterns. The important attributes are gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, InternetService and payment-related details like PaymentMethod, MonthlyCharges and TotalCharges. The main objective of the research is to determine the early churners, which are the customers that terminate the service within the first three months of subscription, because early churn is a major factor in determining business revenue and retention policies.

B. Data Preprocessing

Preprocessing of data was conducted to ensure the quality and usability for machine learning analysis. The TotalCharges column, which was initially stored as a string, was transformed into a numerical format in order to used for performing financial calculations. The cases with missing values were detected and replaced with median strategy. Appropriate encoding of categorical variables was performed: binary variables like Partner, Dependents, and PaperlessBilling are coded as 0 and 1, whereas multi-class variables, such as PaymentMethod, InternetService, Contract, and MultipleLines are one-hot encoded to allow training the model. Gender column was coded 0 Female and 1 Male. The continuous variables (tenure, MonthlyCharges and TotalCharges) were standardized using the StandardScaler to make sure that all numerical variables have the same scale. To overcome the problem of the class imbalance (only 8.48% of the dataset are early churners) the SMOTE (Synthetic Minority Over-sampling Technique) was used, which created artificial examples of the underrepresented group to obtain the balanced dataset.

C. Feature Engineering

Earlychurn target variable was formed by indicating the customers who churned during the first three months (tenure <= 3) to be 1, and other customers 0. This enables the models to consist of early identification of churn that is essential towards the application of appropriate retention measures. No further features were generated for the preprocessing and encoding stage since this research is aimed at testing the standard ML algorithms on the available features.

D. Model Selection and Training.

Early churn prediction was performed using four machine learning models, namely, Logistic Regression, Random Forest, XGBoost, and LightGBM. The baseline linear model was Logistic Regression, and the strong ensemble approach was offered by the Random Forest which can decrease overfitting by bootstrapping and randomizing features. XGBoost and LightGBM have been chosen as gradient boosting models, which have high accuracy, scale, and can work with unbalanced data. Class weighting was used on all models to further explain the minority class. The hyperparameters were optimally configured, and they were the number of estimators, the learning rate, the maximum tree depth, and subsampling ratios of the ensemble models. The data was divided into a training and a testing set in a 75: 25 proportion with stratification to ensure that the class distribution was preserved.

E. Model Evaluation

Several metrics such as precision, recall, F1-score and ROC-AUC were taken to assess model performance as this gives a holistic measure of the effectiveness of classification, particularly when there is imbalance in the classes. Also, the business cost of false positives, false negatives, and true positives was estimated to measure the real-life effect of early churn misclassification. The Precision-Recall curve was used to threshold-optimise the decision boundary and achieve the maximum F1-score to give a trade-off between recall and precision.

F. Model Explainability

SHAP (SHapley Additive exPlanations) values were calculated using the lightGBM model that was selected as the most efficient algorithm in order to increase interpretability. The SHAP analysis can be used to visualize the effect of each feature on the model predictions and identify the main drivers of early churn, including tenure, MonthlyCharges and TotalCharges. This is a step that is crucial to organizations in that they not only get to know who is prone to churn, but also the reasons that would lead to specific intervention strategies.

G. Customer Segmentation

In addition to predictive modeling, KMeans clustering was also performed to segment early churners in terms of tenure, monthly charges, and total charges. The elbow method was used to identify the most appropriate number of clusters, which were three separate segments named Price-Sensitive, Low-Engagement, and Onboarding Failure. These segments have been examined on service usage trends in various features such as OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies. MonthlyCharges were also used to estimate the revenue loss per segment in order to give information on the financial loss of each segment of the churn.

H. Validation

In order to make the models robust, 5-fold stratified cross-validation was carried out in the resampled dataset. The means of the ROC-AUC scores were calculated among folds, which showed consistency and reliability of the predictive models. Another model that was tested was a baseline model with DummyClassifier to demonstrate the benefit of trained machine learning models compared to naive majority-class predictions.

IV. RESULTS

A. Early Churn Rate

The WA_Fn-UseC_-Telco-Customer-Churn dataset analysis revealed that early churn as the number of customers who quit after the first three months of subscription was a rather small yet significant percentage of the customer base. The number of records that were used (n -7,043) identified 597 customers (8.48) as early churners. The fact that most customers stayed with the service even after the first few months but the percentage of early churners highlights the necessity of predictive interventions. It is important to address this segment to limit the loss of revenue as maintaining a current customer is much cheaper than making new customers. The fact that the early churn was low also necessitated the use of oversampling methods like SMOTE to balance the data such that the machine learning models were able to learn and generalize patterns related to early churn events.

B. Descriptive Statistics

In-depth analysis of numeric characteristics allowed gaining an understanding of customer behavior and trends that were linked to early churn. Customer tenure was highly variable with the majority of the customers staying one to thirty months. The first three months were found to be the most important period in which proactive retention strategies have the most significant effect. Monthly Charge Analysis showed that customers that pay more are more likely to leave the service prematurely indicating that cost sensitivity is a significant predictor of early churn. Total Charges that are cumulative with regard to the subscription period naturally rose with tenure, but early churners had low total charges, which were in line with their short-term engagement. Fig 1,2 and 3 present the distributions of tenure, Monthly Charges and Total Charges and show the concentration of early churners at lower tenure and moderate and high monthly payments. These results underscore the need to have models that can capture the relationship among the various characteristics of customers to forecast churn.

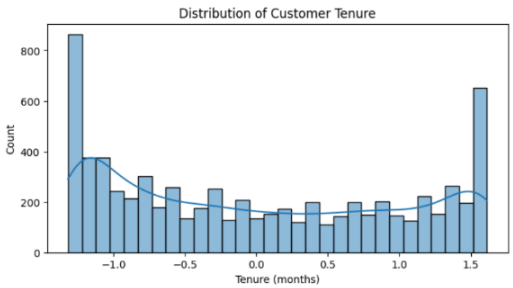

Fig 1:Distribution of customer tenure in the dataset.

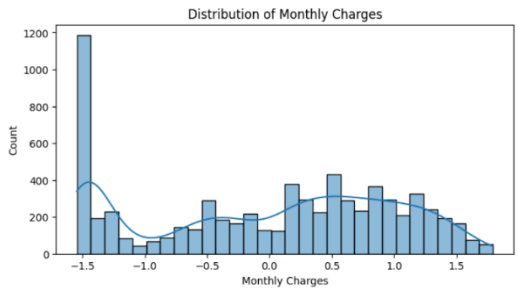

Fig 2: Distribution of monthly charges for all customers.

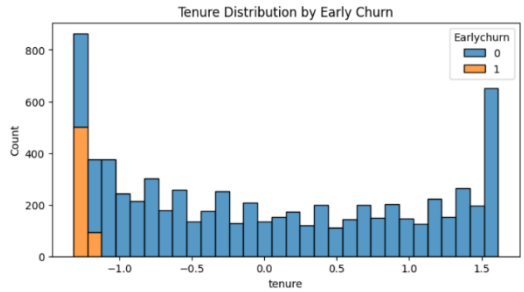

Fig 3: Tenure distribution by early churn status.

C. Performance of the machine learning model.

Four machine learning models namely, Logistic Regression, Random Forest, XGBoost and LightGBM were used to predict early churn. The initial model was the Logistic Regression which was a linear model; the other model was the Random Forest, which is an effective ensemble model and eliminates overfitting through randomizing features and bootstrap aggregation. The choice of gradient boosting models which were deemed to be XGBoost and LightGBM was due to the fact that they are very accurate, can be scaled and can deal with imbalanced data. Table 1: The results of the models on the test set in terms of their performance measures are as follows:

TABLE 1
PERFORMANCE METRICS OF MACHINE LEARNING MODELS FOR EARLY CHURN PREDICTION

Model	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.924	0.997	0.959	0.983
Random Forest	0.962	0.986	0.974	0.994
XGBoost	0.958	0.987	0.972	0.996
LightGBM	0.962	0.984	0.973	0.995

The findings show the ensemble models were far much better than Logistic Regression in accuracy, recall, F1-score, and ROC-AUC. LightGBM was the most successful in terms of overall performance and is computationally efficient, which is why it is the most appropriate model to be deployed in practice. These results are consistent with the existing literature that reveals that ensemble techniques are effective in capturing the intricate patterns of customer behavior and also in reducing the drawbacks of linear models.

D. Threshold Optimization

Threshold optimization was done to strike a balance between precision and recall and business needs using the Precision-Recall curve of the LightGBM model. The decision threshold was set to 0.236, which was the best to recall early churners and false positives were kept at a minimum. At this level, the confusion table shown as table 2 is as follows:

TABLE 2
CONFUSION MATRIX OF LIGHTGBM MODEL AT OPTIMIZED THRESHOLD

	Predicted No	Predicted Yes
Actual No	1,532	80
Actual Yes	17	1,594

Using a business-based cost model whereby the cost of false negatives (FN) is 500, false positives (FP) is 50, and true positives (TP) saves 400, the overall cost at the optimal threshold was found to be -632,400. This was a saving of 14,550 over the default limit (0.5), which emphasizes the utility of threshold tuning in minimizing financial effects and focusing on promoting early retention.

E. Importance and Explainability of Features.

The LightGBM model had the SHAP (SHapley Additive exPlanations) values calculated to identify the most significant features that contribute to early churn. The most important predictor was found to be tenure, and the shorter the tenure the higher the likelihood of early churn. Influence of Monthly Charges was also significant, meaning that the higher the monthly payment is, the higher is the probability of quitting the service. Total Charges was a cumulative engagement metric, and contract type had a strong impact on retention, with month-to-month subscribers being more likely to churn than long-term subscribers did. Additional features of the service, such as TechSupport and OnlineSecurity, also added to predictive insights, which implies that the quality of the service contributes to the influence of financial factors. The results give actionable interpretable information to be used in designing specific retention strategies and intervention programs.

Fig 4 gives the summary plot of the SHAP of the LightGBM model. It brings out the relative contribution of each feature to early churn prediction. Brief tenure, increased amounts of payments per month, and some types of contracts augment the likelihood of early churn. These insights can be used to make specific interventions to high-risk customers.

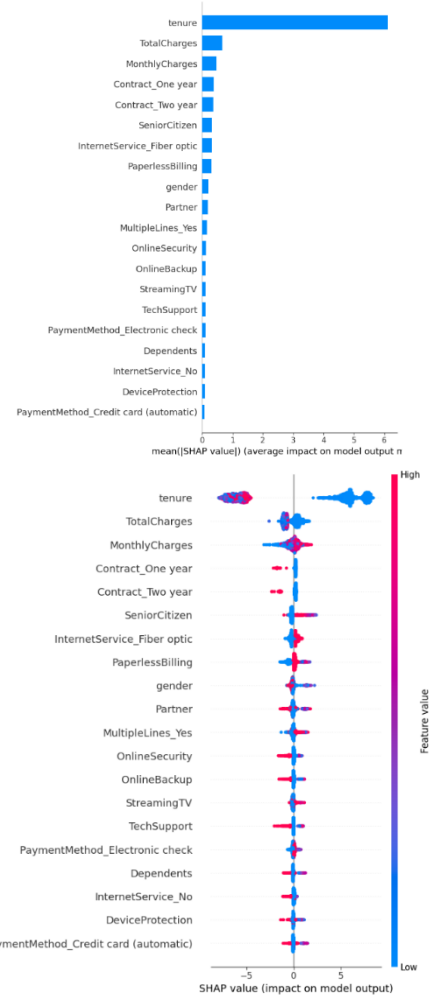

Fig 4: SHAP summary plot showing feature importance for early churn prediction.

F. Customer Segmentation

The KMeans clustering was used on the early churners based on tenure, Monthly Charges, and Total Charges, which formed three different segments. The initial group was Price-Sensitive and covered low spending and short-term customers that showed low interest in value-added services. The second segment Low-Engagement included medium spending and engagement customers. Onboarding Failure was the third segment and was the high-spending customers who churned so early, in many cases because of bad onboarding experiences. The potential financial loss was estimated to be higher in segments, where the Onboarding Failure group was the most contributing segment. Service adoption analysis showed that focused retention programs in this segment would save a significant amount of revenue loss through churn. The segmentation and revenue implications are as shown in fig 5.

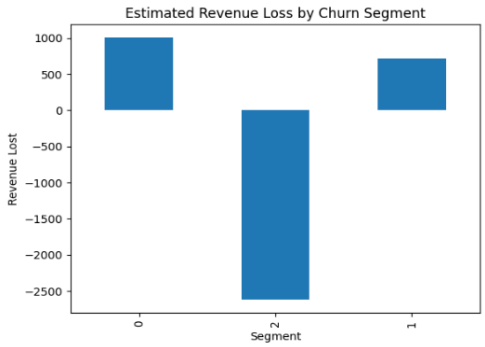

Fig 5: Revenue loss by churn segment.

G. Cross-Validation

In order to guarantee the model robustness, 5-fold stratified cross-validation was conducted on the resampled data. LightGBM model performed well on all folds with the ROC-AUC value of 0.995, 0.998, 0.996, 0.997 and 0.994. The average ROC-AUC was 0.996 and the standard deviation was 0.0013 and this showed that this model is sound and can be generalized. This validation highlights the model ability to work well with hidden data which gives confidence in the implementation of the model to real world churn prediction.

H. Baseline Comparison

A DummyClassifier, which predicts the majority of all the instances, and has ROC-AUC of 0.5 and F1-score of 0.67 in predicting early churners. This comparison goes a long way to show that the trained machine learning model is a significant improvement over the naive models, offering actionable information to business decision-making. The high performance of LightGBM which is enhanced by the optimization of the threshold and understandable insights of the features gives a solid foundation to the targeted retention strategies.

Summary of Key Findings

The analysis has a number of interesting conclusions. Early churn was determined in 8.48 percent of the customer base and the LightGBM model has shown to have the greatest predictive accuracy and ROC-AUC. Optimization at the threshold enhanced the cost of the business and optimized the precision-recall tradeoffs. The major causes of churn were tenure, Monthly Charges, Total Charges and type of contract. Three actionable segments were outlined, providing the

possibilities to implement specific retention interventions. Robustness and generalizability of the model were tested through cross-validation, and the efficacy of machine learning methods compared to naive predictions was tested through baseline comparisons. Together, these findings can give a modern framework upon which early churn in telecommunications can be predicted and alleviated.

V. DISCUSSION

The regression analysis of the WA_Fn-UseC_-Telco-Customer-Churn data gives very useful information about the variables affecting early churn in the telecommunications industry. Early churn, which is the number of the customers who quit within the first three months of subscription, was found to be 8.48% among the customers. Though this shows a relatively small percentage of total customers, the financial impact is great because of the expensive nature of acquiring new customers as opposed to retaining old ones [1].

The researchers compared four machine learning algorithms, including Logistic Regression, Random Forest, XGBoost, and LightGBM, in the context of early churn prediction. The ensemble models, especially LightGBM, always performed better than the Logistic Regression on all the evaluation measures, such as precision, recall, F1-score, and ROC-AUC. These findings affirm previous studies that show better results with ensemble and gradient boosting models in modelling non-linear and complex relationships in customer databases [5], [6]. Although the Logistic Regression offers a simple baseline that is easy to interpret, it does not take into consideration the interactions between two or more features that explain the churn behavior. Conversely, ensemble techniques are useful to handle the interdependencies of features and unbalanced data and are therefore very effective in predicting early churn in a real telecommunication setup [9].

The feature importance analysis using SHAP showed that tenure, Monthly Charges, Total Charges, and contract type are the most influential predictors of early churn. In particular, the customers who had less tenure were much more likely to drop out in the first place as customers are highly sensitive to the first several months of their engagement and retention [10]. The early churn was also related to higher monthly rate and cumulative spending with the result that cost-sensitive customers are highly susceptible to early exit. The type of contract was crucial because month to month subscribers showed more churn propensity than long-term plan customers. Also, features related to service, like TechSupport and OnlineSecurity, were also included in the churn prediction to prove that not only financial, but also qualitative service factors are important in maintaining customer retention [11]. The results are consistent with the literature available that highlights the multifactorial churn behavior, and it involves a combination of monetary and service-based determinants [1], [5], [10].

Segmentation analysis based on KMeans clustering identified three major groups of early churners; Price-Sensitive, Low-Engagement, and Onboarding Failure. Price-Sensitive customers have low spending and low interaction with value-added services, which implies that the cost changes or promotions can enhance retention. Low-Engagement customers show average usage and expenditure which means that it can be beneficial to use loyalty programs or engagement campaigns. The group that has the highest potential loss in revenue is the Onboarding Failure where high-spending customers leave very early.

According to the poor onboarding experience in this group, proactive support, personalized communication, and early engagement initiatives would help reduce the financial losses on churn by a significant margin. These segmentation findings build upon the prior studies by connecting the predictive results to the practical retention interventions, and enable telecom providers to tailor the specific interventions instead of implementing homogenous retention initiatives [9], [11].

The concept of threshold optimization also exhibited the real-world use of predictive modeling. When the LightGBM decision threshold was set to 0.236, instead of the default value of 0.5, the recall of early churners increased and the number of false positives was minimized thus minimizing the potential financial loss that a misclassification would cause. The analysis of the cost, which presupposed FN = 500, FP = 50, and TP = -400, showed the savings of 14,550 in relation to the default threshold. This highlights the need to make predictive models specific to business goals and cost systems because conventional accuracy measures do not portray the realities of operation [10], [11].

The cross-validation performance including a mean ROC-AUC of 0.996 and a very low standard deviation (0.0013) proved the stability and the strength of the LightGBM model when working with different data splits. This brings out the generalizability and reliability of the model to be deployed on invisible customer data, which is essential to operational churn prediction systems [5], [9]. Further experiments with a comparison to a DummyClassifier as the baseline confirmed that machine learning models are able to present actionable information that is much more valuable than a simple majority-class prediction, which justifies the practical value of this method when applied in telecommunications customer retention systems [1].

Overall, the results discussion reveals that an ensemble machine learning models, combined with interpretable feature analysis, threshold optimization, and customer segmentation can offer a holistic and practical framework of early churn predictor. The method fills major gaps in the previous body of research, such as the necessity to have interpretable models, integrate business-relevant metrics, and segmentation-based interventions to combat early churn [5], [9], [10], [11].

VI. CONCLUSION

This paper examined the use of machine learning in churn prediction in the telecommunications industry at the early stages. When the WA_Fn-UseC_-Telco-Customer-Churn dataset was plotted, it was noted that 8.48% of the customers were churning in the initial three months and this indicates the presence of a strategic time in which the retention strategies can be adopted. The performance of four machine learning models, such as the Logistic Regression, Random Forest, XGBoost, and LightGBM, were compared in terms of predictive performance. The accuracy and ROC-AUC of LightGBM are the highest, and the predictive ability and computational efficiency of LightGBM are also high. These

findings are in line with earlier observations that the ensemble techniques are good at capturing the intricate trends in customer behavior [5], [6].

The paper has also shown how optimization of the threshold is critical in balancing precision and recall based on the business priorities. A threshold of 0.236 cut the overall misclassification cost by 14,550 which highlights the practical importance of tuning models beyond default values. The analysis of importance of the features based on SHAP demonstrated that tenure, Monthly Charges, Total Charges, and contract type were the most significant predictors of early churn that allowed taking specific actions to enhance retention interventions. The observations are in line with the previous studies that have suggested that financial and service related issues do affect the churn [1], [10].

Further customer segmentation identified three actionable segments, such as Price-Sensitive, Low-Engagement, and Onboarding Failure, all of which have different patterns of behavior and influence on the revenue. To a great extent, interventions specific to these segments as well as the Onboarding Failure group would help to minimize the possible revenue loss. The predictive models were tested by cross-validation, which confirmed the robustness and generalizability of the models, and allowed using them in real-life contexts.

Altogether, this research paper advances the field by integrating interpretable models of machine learning, cost-sensitive threshold optimization, and segmentation analysis to create an overall framework of early churn prediction. The results highlight the importance of combining predictive analytics with business strategies to enhance customer retention, resource optimization and reduce revenue loss.

VII. LIMITATIONS

Even though this research offers some important findings on the application of classical machine learning models in early churn prediction, it is important to recognize a number of limitations of this study. The analysis has been conducted only using the WA_Fn-UseC_-Telco-Customer-Churn dataset, which is reflective of one telecommunications company only. The applicability of the models to other telecom firms, regions, or industries is therefore not tested and the trends learnt by the models might not be applicable in other customer demographics or service structures [1], [5], [9].

The analysis was done on classical machine learning models only, including Logistic Regression, Random Forest, XGBoost, and LightGBM. These models can be interpreted and are computationally efficient, but might fail to reflect highly non-linear or complex interaction in the behavior of customers, which might be discovered with deep learning methods, including convolutional neural networks or recurrent neural networks [12]. This can restrict predictive accuracy to very large-sized or dynamic datasets.

The second weakness is that the models have been trained with the use of non-evolving historical data that might not capture how the behavior of the customers changes over time. The preferences and churn trends of customers may change with promotions, season, competition in the market or service changes. This is referred to as concept drift which may lower the accuracy of the models in the long term unless this phenomenon is addressed by constant retraining [10].

Even though SHAP analysis was used to offer feature interpretability, such ensemble models as LightGBM and XGBoost remain a black box of certain interactions. This is partly a constraint to the capability of the business stakeholders to comprehend the predictions of the model well and act in a specific manner regarding the interactions of the subtle features [5], [10]. Although threshold optimization was used to minimize the business cost, the other business-important measures like Customer Lifetime Value (CLV), possible revenue impact, and profit optimization were not included. The inclusion of these measures would offer a more detailed assessment of the direct translation of early churn prediction into business results [11].

Lastly, although the dataset contained a range of different demographic, account, and service usage attributes, other potentially critical variables, including customer complaints, social interactions, or sentiment data as service feedback, were not provided. The lack of such behavioral or qualitative data might be a constraint to the richness and accuracy of the predictive models [8], [9].

VIII. FUTURE WORK

Based on the results and the limitations of this research, some areas of further research are proposed to make early churn prediction models more practical and robust. It would be possible to test the robustness and generalizability of the models by expanding the study to more telecom datasets or even cross-industry datasets. This may result in models that are able to support a wide range of customer demographics and service frameworks and enhance applicability across more than one provider [9], [10].

It is possible to examine more complicated and detailed interactions between features by exploring more advanced machine learning methods and deep learning models, such as convolutional neural networks, recurrent neural networks or a hybrid ensemble-deep learning model. This is able to enhance the accuracy of prediction, especially when dealing with large datasets that have complex patterns in customer behavior [12], [6].

In the future, business-focused indicators (Customer Lifetime Value (CLV), increase in net revenue, or profitability of retention interventions) should be included. The use of these metrics as assessment of models would have actionable insights to the management and would directly relate predictive performance to the financial results [11].

Concept drift could be resolved by implementing real-time prediction or periodically retrained models to update the model with the current customer behavior. This makes the predictive system not obsolete or inaccurate with time in response to the changes in the market trends or services offered [10].

Organizations may refine customer segmentation with the help of other behavioral information, patterns of service usage, and demographics that will enable them to develop more personalized retention strategies. Indicatively, integrating churn prediction and clustering or recommendation systems may focus on interventions on individual groups that have the greatest financial potential [6], [7].

Finally, incorporation of qualitative customer feedback, social sentiment analysis, or history of interaction into predictive models may improve the capacity of forecasting early churn. Other sources of data can reveal hidden indicators that cannot be detected by the usual numerical or categorical characteristics and enhance the interpretability and predictive capabilities of the models [8], [9].

By tackling these aspects in the future work, it would increase the accuracy of predictions, increase interpretability and give actionable information to the decision-makers, which would eventually improve customer retention strategies within the telecommunications industry.

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